Synopsis

Air Quality Level Classification System

Problem Id: 19

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- **Problem Description:**

Air pollution is increasing rapidly due to industrialization, vehicles, and urban development. Continuous monitoring of air quality using sensors generates large amounts of environmental data. However, analyzing this data manually is difficult.

The problem is to develop a machine learning model that can classify air quality levels based on sensor readings such as CO, NO_2, temperature, and humidity. The system should preprocess the data, select relevant features, and apply classification algorithms to accurately determine air quality.

Air pollution is one of the major environmental issues affecting human health and ecosystems. Monitoring and predicting air quality helps authorities take preventive actions. This project proposes a machine learning based Air Quality Level Classification System that predicts air quality levels using sensor data such as CO, NO_2, temperature, and humidity.

The dataset is preprocessed to remove noise, handle missing values, and normalize sensor readings. Important features are selected to improve model efficiency. Two machine learning algorithms, Logistic Regression and K-Nearest Neighbors (KNN), are used to classify air quality levels.

The models are evaluated using accuracy and precision metrics to determine their performance.

- **Steps of Implementation:**

1. Data Collection:

First, we will gather an air quality dataset that includes parameters like CO, NO_2, temperature, and humidity. This data can be taken from platforms like Kaggle or other open data sources.

Feature Name	Description	Data Type	Possible Values
StationId	Unique ID of monitoring station	Categorical	AP001, AP002, etc.
Datetime	Date & time of observation	Datetime	Dates
PM2.5	Fine particulate matter concentration	Float	0 – 500 µg/m³
PM10	Coarse particulate matter concentration	Float	0 – 600 µg/m³
NO	Nitric Oxide concentration	Float	0 – 200 µg/m³
NO2	Nitrogen Dioxide concentration	Float	0 – 200 µg/m³
NOx	Nitrogen Oxides concentration	Float	0 – 300 µg/m³
NH3	Ammonia concentration	Float	0 – 200 µg/m³
CO	Carbon Monoxide concentration	Float	0.1 – 10.0 ppm
SO2	Sulfur Dioxide concentration	Float	0 – 150 µg/m³
O3	Ozone concentration	Float	0 – 180 µg/m³
Benzene	Benzene concentration	Float	0 – 50 µg/m³
Toluene	Toluene concentration	Float	0 – 100 µg/m³
Xylene	Xylene concentration	Float	0 – 100 µg/m³
AQI	Air Quality Index value	Integer	0 – 500
AQI_Bucket	AQI category (target variable)	Categorical	Good, Satisfactory, Moderate, Poor, Very Poor, Severe

2. Data Preprocessing:

Next, we will clean the data by handling missing values and removing unwanted noise. We will also scale the data so that all values are in a similar range, which helps the model perform better.

To improve model performance, the following preprocessing steps will be applied:

- **Handling Missing Values**
  - Use mean/median imputation for numerical data
  - Use mode for categorical data
- **Noise Removal**
  - Remove outliers using IQR (Interquartile Range) or Z-score method

    Z = (x-u) / o (u is mean, o is standard deviation)

- **Feature Scaling**
    - Apply Min-Max Scaling or Standardization (Z-score normalization)

      Min -Max Scaling : Xnorm = (X-Xmin) / (Xmax-Xmin)

      X-Standard : Xstd = (X-u) / o (u is mean, o is standard deviation)
    - Important for distance-based models like KNN
- **Encoding Categorical Data**
    - Convert wind direction or AQI categories using Label Encoding / One-Hot Encoding
- **Feature Selection**
    - Use correlation matrix or feature importance to remove less useful features

3. Model Selection:

We will use the following algorithms:

K-Nearest Neighbors (KNN)

- Works well for classification problems
- Uses distance between data points
- Suitable because air quality data has continuous numerical features

Logistic Regression

- Simple and efficient classification algorithm
- Provides probabilistic outputs
- Works well when relationship is linear

4. Training and Validation:

- The dataset will be split into:

  Training Set (70–80%)

  Testing Set (20–30%)
- Training Process:

  Models (KNN & Logistic Regression) will be trained using training data
- Validation Techniques:

  Train-Test Split Method

  Cross-Validation for better reliability
- Hyperparameter tuning:

  K value in KNN

  Regularization in Logistic Regression

5. Evaluation:

The performance of models will be evaluated using:

- **Accuracy** : Overall correctness
- **Precision** : Correct positive predictions
- **Confusion Matrix** : Detailed classification results

6. Final Model:

Finally, we will select the model that gives the best results and use it for predicting air quality levels.

➢ Expected Modules/ Libraries to be used :

- **NumPy:**

It is used to perform numerical calculations and work with arrays efficiently, which makes handling large amounts of data easier.

- **Pandas:**

It helps in organizing and managing the dataset, making it simple to read, clean, and analyze data.

- **Matplotlib:**

It is used to create basic graphs and charts, helping us visualize the data in a clear way.

- **Seaborn:**

It provides more advanced and attractive visualizations, such as heatmaps, to better understand relationships in the data.

- **Scikit-learn (sklearn):**

It is the main library used to build machine learning models like KNN and Logistic Regression, and also to evaluate their performance using different metrics.

➢ Expected Evaluation metrics and Plots to be used:

1. Accuracy: Measures overall correctness of the model.

2.Precision: Measures correctness of positive predictions.

3.Confusion Matrix: Shows correct and incorrect classifications.

4.Heatmap: Visual representation of correlation between features.

5.Bar Graphs: Used to compare model performance.

These metrics and plots help in understanding model performance and selecting the best model.

- **Required topics from the subject:**

- **Feature Selection:**

Feature Selection is used to identify and choose the most relevant input variables from the dataset that contribute the most to the prediction. This helps in reducing unnecessary data, improving model performance, and decreasing computational complexity.

- **Logistic Regression:**

Logistic Regression is a supervised machine learning algorithm used for classification problems. It predicts the probability of a data point belonging to a particular class by using a sigmoid function, making it useful for binary outcomes like yes/no or 0/1 predictions.

- **K-Nearest Neighbors (KNN):**

K-Nearest Neighbors (KNN) is a simple, instance-based learning algorithm used for classification and regression. It works by finding the 'K' closest data points to a given input and assigning the most common class among those neighbors.

- **Platform Required:**

1.Jupyter Notebook

2.Python IDLE

3.Visual Studio Code

- **Books and Link Sources:**

1. "Hands-On Machine Learning with Scikit-Learn" by Aurélien Géron

2. Python for Data Analysis" by Wes McKinney

3. Kaggle (https://www.kaggle.com/□)

4. Scikit-learn Documentation (https://scikit-learn.org/□)